LIRUI WANG

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EDUCATION

University of Washington, Seattle *Sep 2016 - Mar 2021*
Bachelor of Science, Electrical Engineering (Honors), College of Engineering
Bachelor of Science, Computer Science (Honors), College of Arts and Sciences
Bachelor of Science, Mathematics (Minor), College of Arts and Sciences
Master of Science, Computer Science, College of Arts and Sciences
GPA: 3.96/4.00, *summa cum laude*
Advisor: Professor Dieter Fox

RESEARCH PROJECTS

6D Instance Grasping with Point Cloud *Jan 2020 - Sep 2020*

- Investigated actor-critics methods with planner demonstrations and goal auxiliary tasks, to learn closed-loop control policies for 6D robotic grasping using point clouds.
- Achieved over 90% success rates on grasping various unseen objects in both simulation and real world.

Joint Motion and Grasp Planning *Mar 2019 - Dec 2019*

- Proposed an optimization-based integrated planner that optimizes manipulation trajectory with online grasp synthesis and selection.
- Demonstrated robust and smooth motion plans for grasping in cluttered scenes in the real world and outperformed several fixed-endpoint baselines.

Pose Tracking with the Deep Iterative Matching Network *Jun 2018 - Mar 2019*

- Evaluated DeepIM on YCB Video dataset and improved training on synthetic data by generating diverse pose distribution and realistic occlusions.
- Extended DeepIM to achieve state-of-the-art pose tracking accuracy and AR applications.

Unsupervised Flow and Depth Methods Study *Mar 2018 - Jun 2018*

- Studied structure from motion methods that jointly learn 3D vision tasks such as depth and flow.
- Explored self-supervised learning works such as SfMlearner and MonoDepth and analyzed in a static scene setting.

Baxter Hand-Eye Coordination *Sep 2017 - Mar 2018*

- Developed software for the Baxter robot to grasp and manipulate objects detected by PoseCNN.
- Studied and implemented camera models, hand-eye calibrations, robot kinematics and dynamics, shader-based rendering and different ROS packages for tracking and control.

WORK EXPERIENCE

Seattle, WA *Jun 2020 - Oct 2020*
Applied Research Intern, NVIDIA

- Investigated model-based RL for autonomous driving with a multi-modal RNN traffic predictive model from the bird's eye view.

- Used the predictive model latent to simulate reactive traffic and to learn adaptive cruise control with cut-in avoidance and adopted the learned model for long-horizon motion planning.

Seattle, WA *Sep 2019 - Present*
Teaching Assistant, UW CSE Department

- Assisted teaching for computer vision, deep learning, and artificial intelligence courses at UW CSE.
- Held office hours, helped and graded homework, and prepared course materials.

Shenzhen, China *Jun 2017 - Aug 2017*
Robotics Engineer Intern, DJI Technology

- Implemented functions such as take-off, tag detection, and multi-robot communication for drones.
- Applied SLAM to mapping and localization, and built a state machine to generate high-level plans.

LEADERSHIP AND INVOLVEMENT

Seattle, WA *Sep 2016 - Feb 2018*
ARUW Computer Vision Lead

- Managed a team of 10 students to design computer vision algorithms for an autonomous robot battle, including robots with onboard NVIDIA TX1 and drones with DJI Manifold with Guidance, in the RoboMasters competition hosted by DJI.
- Performed system integration of feedback controls, navigation systems, and serial communication.

MISCELLANEOUS PROJECTS

- **Spatial Audio Processing in the SoundScape System**: Designed a sound processing pipeline from scratch to model the acoustics of a train whistle.
- **Sheet Music Editor**: Created a demo app that can record sound pieces and translate into sheet music display UI using DTFT.
- **Embedded Drowsiness Detection System**: Built an embedded system on Raspberry Pi that prototypes a real-time drowsiness detection system based on human face keypoint detection.
- **Survey and Empirical Evaluation of Attacks and Defense for Robust Classification**: Investigated example meta-reweighting and adversarial training techniques to robustify a fossil classification task and surveyed on attacks and defense of adversarial examples for neural networks.
- **Continuous Trajectory Optimization with Machine Learning Techniques**: Studied formulation of trajectory optimization as functional gradient descent using Reproducing Kernel Hilbert Space and as probabilistic inference on a factor graph using Gaussian Process.
- **Microsoft Server Mover Project**: Built an autonomous mobile manipulator, a UR5 arm mounted on a server forklift, to pull out servers from a server rack.

HONORS

- Annual Deans List, University of Washington, *2016 - 2020*
- Tau Beta Pi Scholarship *2017 - 2019*
- Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship *2017 - 2018*
- Patricia Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship *2018 - 2019*
- Herbold Data Science Fellowship *2019 - 2020*

PUBLICATIONS

- **Goal-Auxiliary Actor-Critic for 6D Robotic Grasping with Point Clouds**
  Lirui Wang, Yu Xiang, Dieter Fox
  Under Review
- **Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection**
  Lirui Wang, Yu Xiang, Dieter Fox
  In Proceedings of Robotics: Science and Systems (RSS), 2020

SELECTED COURSES

- **Computer Science**: Data Structure and Parallelism, Algorithms, Computer Vision, Artificial Intelligence, Deep Learning, Modern Algorithm Toolbox, Graphics, Mobile Robots, Robustness in Machine Learning, Interactive Learning, Probabilistic Robotics, Reinforcement Learning, Data Science
- **Electrical Engineering**: Digital Signal Processing and Filtering, Control Theory, Embedded System, Discrete Time Linear System, Control Capstone, Optimization and Learning for Control
- **Math**: Multivariate Calculus, Probability and Statistics, Numerical Analysis, Linear Analysis, Topology, Convex Optimization and Analysis, High Dimensional Statistics, Statistical Learning
- **English**: Technical Writing, Advanced Technical Writing in Electrical Engineering

SKILLS

- **Platforms**: Arduino, NVIDIA TX Series, STM32, Franka Panda, Baxter, UR5e, DJI Manifold
- **Languages**: Python, C++/C, Matlab, Java
- **Tools**: Git, CMake, OpenCV, Tensorflow, Pytorch, Latex, OpenGL, ROS, Ubuntu
- **Hardware**: 3D Printing, Laser Cutting, Machining, Soldering